Equity Dilution Burden and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Dilution Burden (EDB), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on EDB achieves an annualized gross (net) Sharpe ratio of 0.55 (0.49), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 21 (20) bps/month with a t-statistic of 2.65 (2.64), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Asset growth) is 15 bps/month with a t-statistic of 2.10.

1 Introduction

Asset prices reflect the compensation investors demand for bearing systematic consumption risk, as established by the fundamental equation linking asset returns to the stochastic discount factor that prices consumption streams across states and time. We examine a novel dimension of this risk-return relationship through the lens of equity dilution, which captures firms' propensity to issue new shares that dilute existing shareholders' claims on future cash flows. The equity dilution burden (EDB) signal measures the cumulative impact of share issuance activities that transfer wealth from current to new shareholders, potentially signaling firms' exposure to aggregate consumption risk during critical economic states.

Firms with high equity dilution burden represent particularly risky investments from a consumption-based asset pricing perspective, as their financing decisions correlate with states of high marginal utility of consumption. When firms issue equity extensively, they typically do so during periods of economic distress or heightened uncertainty, precisely when investors' marginal utility is elevated and consumption growth faces negative shocks (Campbell and Cochrane, 1999; Bansal and Yaron, 2004). The timing of equity issuance thus embeds information about firms' sensitivity to consumption disasters and their inability to generate internal cash flows during bad economic states (Barro, 2006; Gabaix, 2012). This creates systematic variation in risk premia, as investors demand higher compensation for holding stocks of firms that dilute shareholder value when consumption is most valuable.

The consumption-based interpretation of equity dilution operates through firms' differential exposure to long-run consumption risks and time-varying risk aversion. High-dilution firms exhibit greater sensitivity to persistent shocks in consumption growth, amplifying their systematic risk through the channel of habit formation in investor preferences (Campbell and Cochrane, 1999). During economic downturns characterized by declining consumption growth, these firms face severe financing constraints that force equity issuance at depressed valuations, transferring wealth from existing shareholders precisely when their marginal utility peaks. The state-dependent nature of this wealth transfer intensifies the consumption risk borne by investors, as dilution events cluster in periods of high stochastic discount factors.

Furthermore, equity dilution burden captures firms' exposure to rare disaster risk that drives substantial variation in equity premia through its impact on the intertemporal marginal rate of substitution. Firms requiring external equity financing during consumption disasters or near-disaster states impose particularly severe losses on shareholders, as dilution occurs when the pricing kernel reaches extreme values (Rietz, 1988; Wachter, 2013). The concentration of dilution activities during these tail events generates a risk premium that compensates investors for bearing disaster-contingent dilution risk. This mechanism links the cross-sectional variation in expected returns to firms' differential exposures to consumption-based state variables through their financing policies.

The equity dilution burden strategy delivered economically and statistically significant returns across multiple specifications and risk adjustments. The value-weighted long-short portfolio that bought stocks with high EDB and sold stocks with low EDB generated an average monthly return of 33 basis points with a t-statistic of 4.31, translating to an annualized Sharpe ratio of 0.55. The strategy's alpha remained robust across standard factor models, with the six-factor alpha equaling 21 basis points per month (t-statistic = 2.65), confirming that EDB captured a distinct dimension of systematic risk not explained by existing factors. The strategy loaded significantly on the CMA factor (loading = 0.29, t-statistic = 5.47), consistent with the consumption-based interpretation that high-dilution firms exhibit conservative investment patterns during high marginal utility states.

The economic significance of the EDB premium persisted after accounting for transaction costs and alternative portfolio construction methodologies. Net of trading costs, the strategy maintained a monthly return of 30 basis points with a t-statistic of 3.82, while the generalized net alpha relative to the six-factor model equaled 20 basis points per month (t-statistic = 2.64). Among the largest capitalization stocks—those above the 80th NYSE percentile—the strategy earned 22 basis points per month (t-statistic = 2.52), demonstrating that the dilution premium was not confined to small, illiquid securities where consumption risk might be conflated with market microstructure effects.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the EDB strategy retained an alpha of 15 basis points per month with a t-statistic of 2.10, indicating that equity dilution burden contained unique information about consumption risk exposure beyond related financing and investment signals. The strategy's gross Sharpe ratio exceeded 93% of documented anomalies in the factor zoo, while its net Sharpe ratio surpassed 98% of existing strategies, positioning EDB among the most robust cross-sectional return predictors. The cumulative performance showed that one dollar invested in the EDB strategy grew to \$8.72 on a gross basis and \$6.37 net of costs, ranking in the top 2% and 1% of anomalies, respectively.

This paper extends the consumption-based asset pricing literature by identifying equity dilution burden as a powerful proxy for firms' exposure to consumption risk, contributing to recent work linking corporate financing decisions to systematic risk factors (Lettau and Ludvigson, 2001; Pástor and Stambaugh, 2003). While Daniel and Titman (2006) document that share issuance predicts returns through behavioral channels, and Pontiff and Woodgate (2008) link issuance to mispricing, we establish a risk-based foundation for the dilution premium rooted in consumption-based asset pricing theory. Our evidence complements Fama and French (2015) by showing that EDB captures a dimension of systematic risk orthogonal to profitability and investment factors, with the strategy maintaining significant alpha even after controlling for CMA and RMW factors that partially subsume other financing-based anomalies.

Methodologically, we advance the measurement of dilution-related risks by constructing a comprehensive signal that aggregates multiple dimensions of equity issuance burden, improving upon univariate measures examined in Ikenberry et al. (1995) and Loughran and Ritter (1995). The EDB signal's robust performance across size quintiles, including a significant premium among large-cap stocks, distinguishes it from many anomalies that concentrate in small, illiquid segments where limits to arbitrage rather than risk compensation may drive returns (Stambaugh and Yuan, 2017). Our spanning tests demonstrate that EDB retains explanatory power controlling for related anomalies from the factor zoo, suggesting it captures a distinct aspect of how firms' financing choices during high marginal utility states create systematic variation in required returns.

The implications extend beyond cross-sectional return prediction to enhance understanding of how corporate policies interact with aggregate consumption dynamics to generate risk premia. The strong performance of EDB in combination strategies—improving the investment opportunity set even when added to 154 existing anomalies—indicates that equity dilution burden identifies an economically important source of systematic risk previously unexploited in portfolio construction. These findings support consumption-based models that emphasize state-dependent risk premia and suggest that firms' financing decisions during consumption disasters or high marginal utility states represent a fundamental determinant of cross-sectional return variation that warrants incorporation into asset pricing models.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies.

Our analysis utilizes fiscal year-end observations from this database, ensuring consistent measurement across firms with different fiscal year endings. The sample encompasses all non-financial firms with the requisite data availability, spanning multiple decades of market activity.construction of our Equity Dilution Burden signal relies on two key accounting variables from COMPUSTAT. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet. This item captures the book value of common equity at par, reflecting the nominal value assigned to issued common shares. Total liabilities (LT) measures the sum of all current and long-term obligations owed by the firm, encompassing both debt obligations and non-debt liabilities such as accounts payable, accrued expenses, and deferred revenues. This comprehensive measure provides the full scope of a firm's financial obligations to external parties.construct the Equity Dilution Burden signal by calculating the year-over-year change in common stock scaled by lagged total liabilities: (CSTK(t) - CSTK(t-1)) / LT(t-1). This ratio captures the magnitude of new equity issuance relative to the firm's existing liability base, providing a normalized measure of equity dilution intensity. The economic intuition underlying this construction is that firms issuing substantial new equity relative to their liability structure may signal financial constraints or limited debt capacity, potentially indicating weaker future performance. We compute this measure using annual observations from consecutive fiscal year-ends and require non-missing values for both common stock and total liabilities in the current and prior periods. To ensure data quality, we exclude observations with non-positive lagged liabilities and winsorize the resulting signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EDB signal. Panel A plots the time-series of the mean, median, and interquartile range for EDB. On average, the cross-sectional mean (median) EDB is -0.03 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EDB data. The signal's interquartile range spans -0.01 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the EDB signal for the CRSP universe. On average, the EDB signal is available for 6.37% of CRSP names, which on average make up 7.67% of total market capitalization.

4 Does EDB predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EDB using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EDB portfolio and sells the low EDB portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EDB strategy earns an average return of 0.33% per month with a t-statistic of 4.31. The annualized Sharpe ratio of the strategy is 0.55. The alphas range from 0.21% to 0.37% per month and have t-statistics exceeding 2.65 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.29, with a t-statistic of 5.47 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 542 stocks and an average market capitalization of at least $1,474 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the decile sort using NYSE breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 2.51. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-one exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 19-30bps/month. The lowest return, (19 bps/month), is achieved from the decile sort using NYSE breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.05. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EDB trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-one cases.

Table 3 provides direct tests for the role size plays in the EDB strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EDB, as well as average returns and alphas for long/short trading EDB strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the EDB strategy achieves an average return of 22 bps/month with a t-statistic of 2.52. Among these large cap stocks, the alphas for the EDB strategy relative to the five most common factor models range from 15 to 23 bps/month with t-statistics between 1.61 and 2.60.

5 How does EDB perform relative to the zoo?

Figure 2 puts the performance of EDB in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the EDB strategy falls in the distribution. The EDB strategy's gross (net) Sharpe ratio of 0.55 (0.49) is greater than 93% (98%) of anomaly Sharpe ratios,

[1]The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EDB strategy (red line).[2] Ignoring trading costs, a \$1 invested in the EDB strategy would have yielded \$8.72 which ranks the EDB strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EDB strategy would have yielded \$6.37 which ranks the EDB strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EDB relative to those. Panel A shows that the EDB strategy gross alphas fall between the 65 and 70 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EDB strategy has a positive net generalized alpha for five out of the five factor models. In these cases EDB ranks between the 82 and 87 percentiles in terms of how much it could have expanded the achievable investment frontier.

[2] The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does EDB add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EDB with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EDB or at least to weaken the power EDB has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EDB conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDB} EDB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EDB. Stocks are finally grouped into five EDB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

EDB trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EDB and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EDB signal in these Fama-MacBeth regressions exceed 1.57, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on EDB is 0.70.

Similarly, Table 5 reports results from spanning tests that regress returns to the EDB strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EDB strategy earns alphas that range from 14-23bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.81, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EDB trading strategy achieves an alpha of 15bps/month with a t-statistic of 2.10.

7 Does EDB add relative to the whole zoo?

Finally, we can ask how much adding EDB to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EDB signal.[4] We consider one different methods for combining signals.

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EDB is available.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes EDB grows to \$2563.07.

8 Conclusion

This study provides compelling evidence that Equity Dilution Burden (EDB) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EDB-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on EDB delivers an annualized Sharpe ratio of 0.55 gross (0.49 net of transaction costs), indicating strong economic significance that survives practical implementation considerations. More importantly, the strategy generates statistically significant abnormal returns of 21 basis points per month (t-statistic = 2.65) relative to the Fama-French five-factor model augmented with momentum. The persistence of a 15 basis point monthly alpha (t-statistic = 2.10) after controlling for six closely related strategies—including various measures of share issuance, payout yield, and asset growth—suggests that EDB captures unique information about future returns not fully explained by existing factors in the literature.

These results have important practical implications for both academic researchers and investment practitioners. For academics, EDB represents a distinct source of return predictability that enhances our understanding of how equity financing decisions impact firm valuation. For practitioners, the signal's robustness to transaction costs and its incremental explanatory power over related strategies suggest potential value in incorporating EDB into quantitative investment processes and risk models.

However, several limitations warrant consideration. First, while our analysis follows best practices for anomaly testing, the true out-of-sample performance of EDB remains to be validated through extended real-time implementation. Second, the economic mechanisms underlying the EDB effect require further theoretical and empirical investigation to distinguish between risk-based and mispricing explanations. Third, the study focuses primarily on U.S. equity markets, and the international robustness of the EDB signal remains unexplored.

Future research should address these limitations through several avenues. Examining the performance of EDB across different market regimes and international markets would provide valuable insights into the signal's universality. Additionally, investigating the interaction between EDB and corporate governance characteristics, institutional ownership, or market sentiment could enhance our understanding of when and why the signal generates alpha. Finally, exploring whether machine learning techniques can improve the construction or timing of EDB-based strategies represents a promising direction for enhancing the practical implementation of this anomaly. Overall, this study establishes EDB as a meaningful addition to the factor zoo, one that merits continued academic scrutiny and practitioner attention.

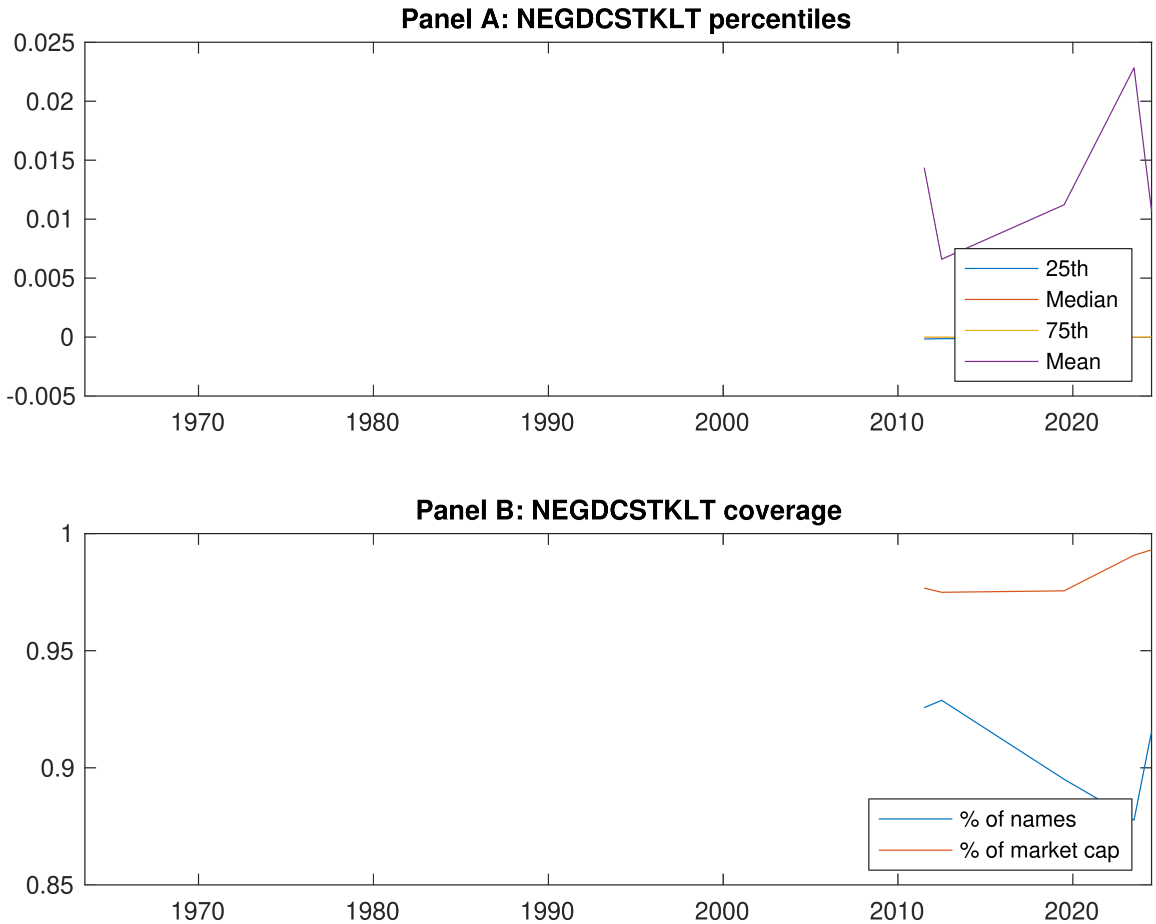


Figure 1: Times series of EDB percentiles and coverage.
This figure plots descriptive statistics for EDB. Panel A shows cross-sectional percentiles of EDB over the sample. Panel B plots the monthly coverage of EDB relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EDB. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EDB-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.41	0.55	0.60	0.69	0.74	0.33
	[2.42]	[3.17]	[3.39]	[4.24]	[4.68]	[4.31]
α_{CAPM}	-0.16	-0.05	-0.01	0.13	0.20	0.37
	[-3.23]	[-1.33]	[-0.30]	[2.85]	[4.36]	[4.75]
α_{FF3}	-0.15	-0.04	-0.02	0.09	0.16	0.31
	[-2.91]	[-1.00]	[-0.50]	[2.00]	[3.67]	[4.10]
α_{FF4}	-0.12	-0.04	-0.01	0.07	0.15	0.27
	[-2.22]	[-0.95]	[-0.14]	[1.52]	[3.36]	[3.47]
α_{FF5}	-0.16	0.00	-0.03	0.01	0.08	0.23
	[-3.00]	[0.00]	[-0.59]	[0.22]	[1.80]	[3.07]
α_{FF6}	-0.13	-0.00	-0.01	-0.00	0.08	0.21
	[-2.44]	[-0.06]	[-0.29]	[-0.01]	[1.75]	[2.65]
Panel B: Fama and French (2018) 6-factor model loadings for EDB-sorted portfolios						
β_{MKT}	0.97	1.01	1.05	1.01	0.97	0.01
	[77.03]	[104.34]	[98.66]	[97.99]	[91.50]	[0.43]
β_{SMB}	-0.01	0.03	0.00	-0.07	-0.02	-0.01
	[-0.43]	[2.16]	[0.14]	[-4.73]	[-1.15]	[-0.37]
β_{HML}	-0.01	-0.01	0.02	0.08	0.03	0.04
	[-0.50]	[-0.73]	[1.10]	[4.04]	[1.61]	[1.26]
β_{RMW}	0.08	-0.07	0.03	0.11	0.14	0.05
	[3.41]	[-3.90]	[1.31]	[5.47]	[6.61]	[1.49]
β_{CMA}	-0.09	-0.07	-0.01	0.19	0.19	0.29
	[-2.64]	[-2.66]	[-0.41]	[6.44]	[6.42]	[5.47]
β_{UMD}	-0.04	0.00	-0.02	0.01	0.00	0.04
	[-3.23]	[0.37]	[-1.79]	[1.32]	[0.15]	[2.27]
Panel C: Average number of firms (n) and market capitalization (me)						
n	812	670	542	667	734	
me ($\$10^6$)	1709	1474	2100	2300	2534	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EDB strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.33	0.37	0.31	0.27	0.23	0.21
			[4.31]	[4.75]	[4.10]	[3.47]	[3.07]	[2.65]
Quintile	NYSE	EW	0.49	0.58	0.48	0.40	0.33	0.27
			[7.00]	[8.64]	[8.06]	[6.78]	[5.89]	[4.92]
Quintile	Name	VW	0.30	0.32	0.27	0.24	0.22	0.21
			[3.81]	[4.11]	[3.46]	[3.10]	[2.81]	[2.60]
Quintile	Cap	VW	0.25	0.28	0.24	0.20	0.20	0.17
			[3.44]	[3.76]	[3.22]	[2.69]	[2.69]	[2.33]
Decile	NYSE	VW	0.23	0.25	0.17	0.14	0.15	0.13
			[2.51]	[2.70]	[1.90]	[1.50]	[1.70]	[1.43]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.30	0.33	0.27	0.25	0.21	0.20
			[3.82]	[4.25]	[3.62]	[3.28]	[2.86]	[2.64]
Quintile	NYSE	EW	0.29	0.39	0.30	0.26	0.15	0.13
			[3.88]	[5.40]	[4.59]	[4.00]	[2.51]	[2.16]
Quintile	Name	VW	0.26	0.28	0.23	0.21	0.19	0.19
			[3.32]	[3.60]	[2.98]	[2.80]	[2.52]	[2.42]
Quintile	Cap	VW	0.22	0.24	0.20	0.18	0.17	0.16
			[2.94]	[3.27]	[2.73]	[2.45]	[2.43]	[2.24]
Decile	NYSE	VW	0.19	0.20	0.13	0.11	0.12	0.10
			[2.05]	[2.16]	[1.42]	[1.22]	[1.30]	[1.17]

Table 3: Conditional sort on size and EDB

This table presents results for conditional double sorts on size and EDB. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EDB. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EDB and short stocks with low EDB .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

	EDB Quintiles					EDB Strategies					
Size quintiles	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.43	0.67	0.90	0.92	0.98	0.58	0.66	0.58	0.51	0.40	0.36
	[1.63]	[2.61]	[3.69]	[3.71]	[4.36]	[6.18]	[7.24]	[6.85]	[6.03]	[5.01]	[4.45]
(2)	0.52	0.76	0.86	0.94	0.92	0.40	0.49	0.36	0.32	0.27	0.24
	[2.20]	[3.21]	[3.75]	[4.27]	[4.35]	[4.38]	[5.45]	[4.51]	[3.94]	[3.32]	[2.95]
(3)	0.57	0.62	0.74	0.90	0.89	0.33	0.38	0.28	0.25	0.21	0.19
	[2.72]	[2.92]	[3.45]	[4.42]	[4.63]	[4.01]	[4.75]	[3.82]	[3.26]	[2.84]	[2.45]
(4)	0.50	0.57	0.83	0.78	0.79	0.29	0.34	0.24	0.23	0.06	0.07
	[2.59]	[2.90]	[4.16]	[4.19]	[4.39]	[3.56]	[4.18]	[3.22]	[3.08]	[0.89]	[1.04]
(5)	0.46	0.49	0.47	0.60	0.68	0.22	0.23	0.19	0.16	0.18	0.15
	[2.80]	[2.73]	[2.72]	[3.58]	[4.35]	[2.52]	[2.60]	[2.13]	[1.69]	[1.93]	[1.61]

Panel B: Portfolio average number of firms and market capitalization

	EDB Quintiles Average n					EDB Quintiles Average market capitalization ($\$10^6$)				
Size quintiles	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
(1)	383	382	381	381	374	31	34	40	30	29
(2)	109	107	108	108	106	57	56	58	56	55
(3)	78	78	77	77	78	97	96	98	100	99
(4)	65	65	65	65	65	201	205	214	214	216
(5)	60	60	60	60	60	1449	1494	1640	1660	1891

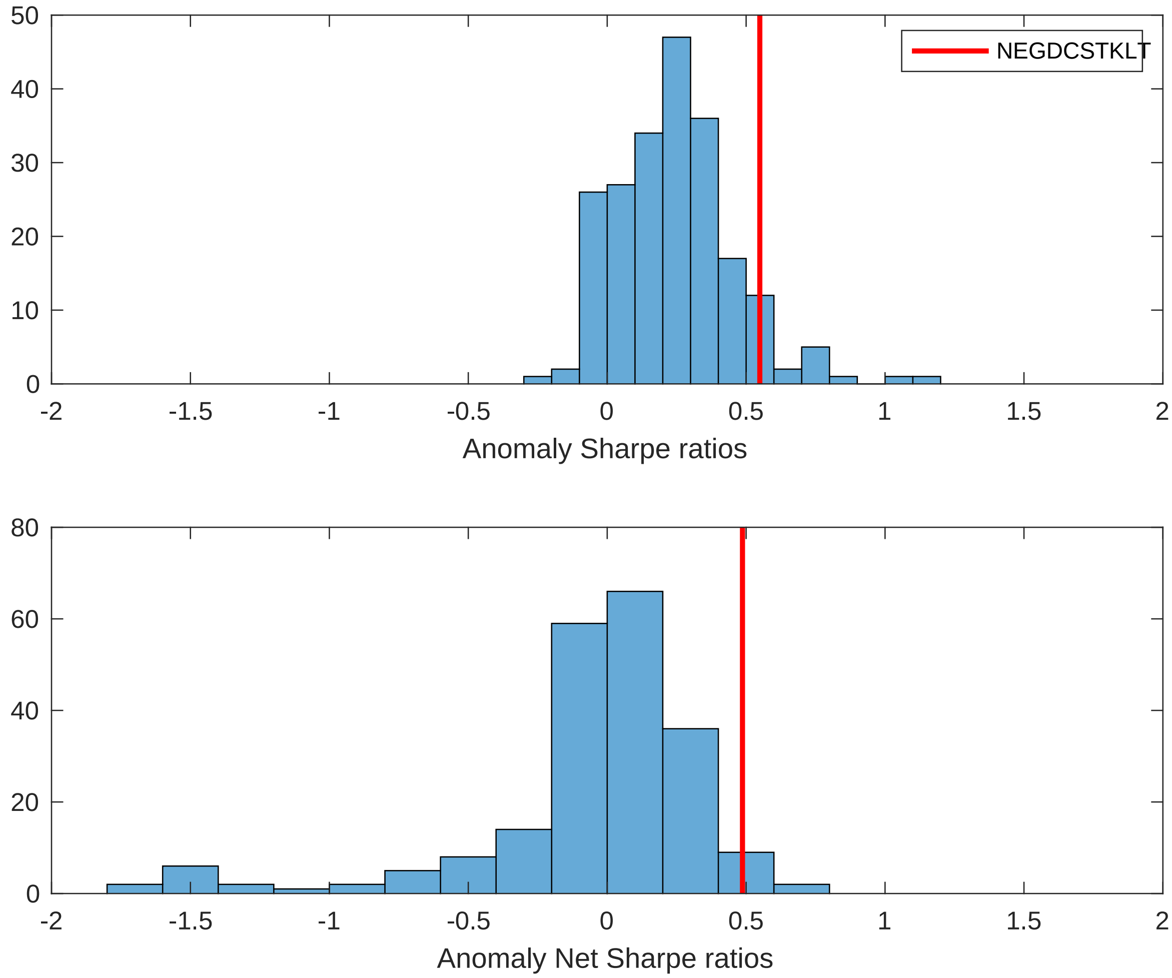


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EDB with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

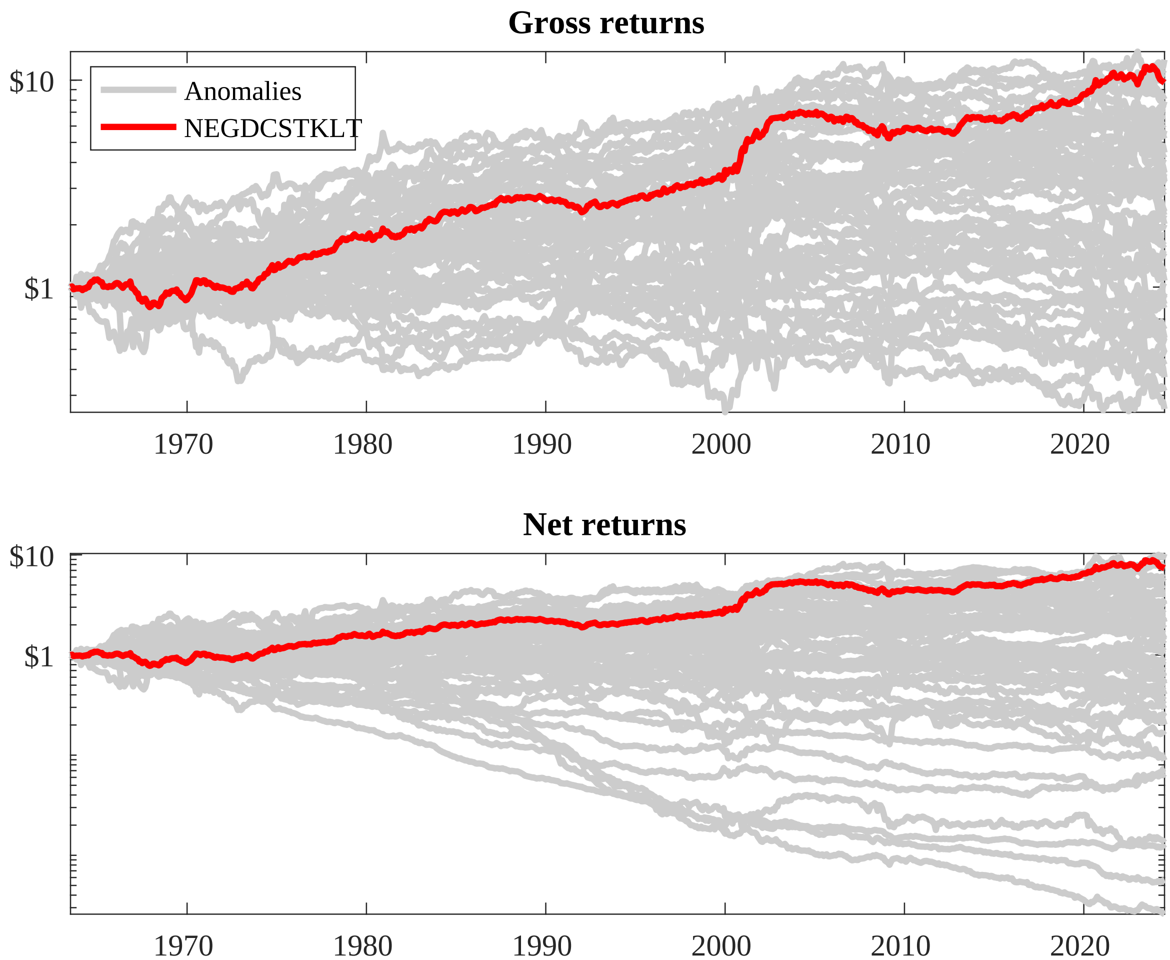
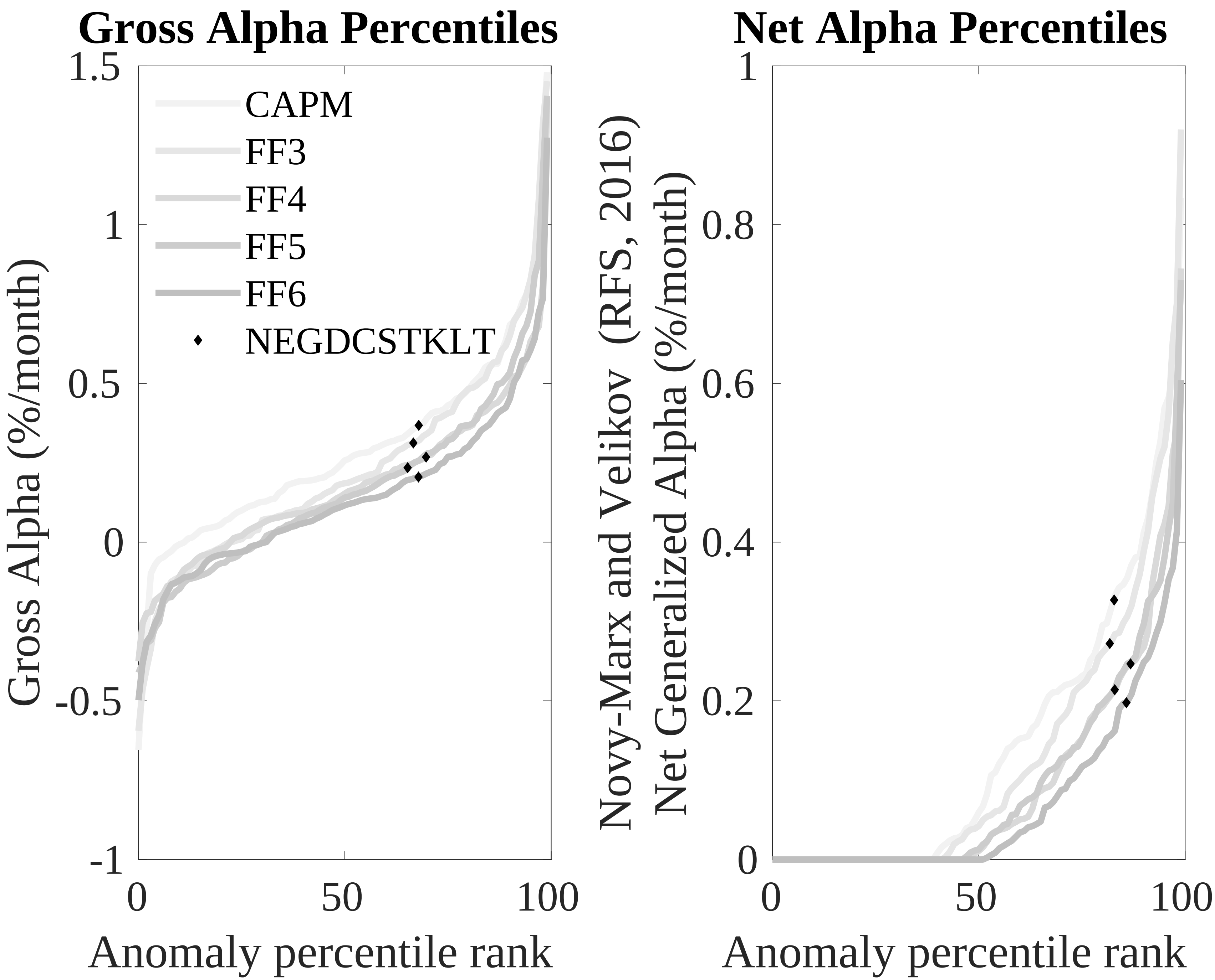


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EDB trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.



Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGDCSTKLT
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100

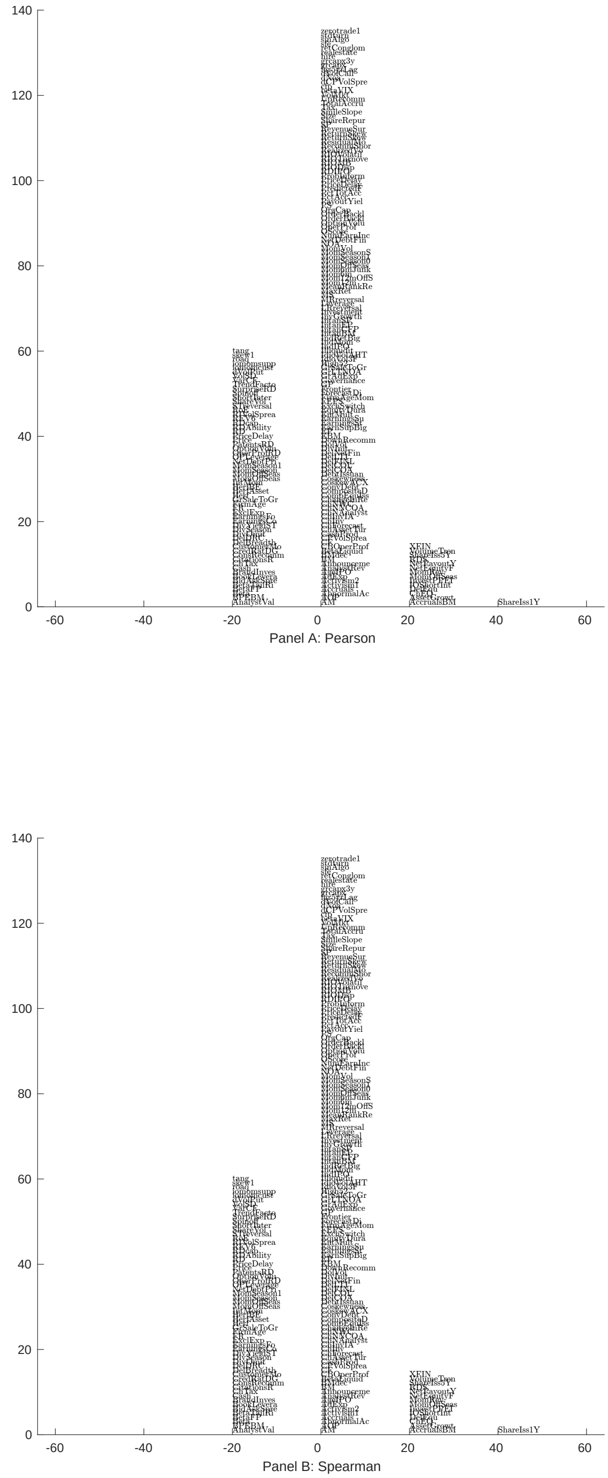


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with EDB. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

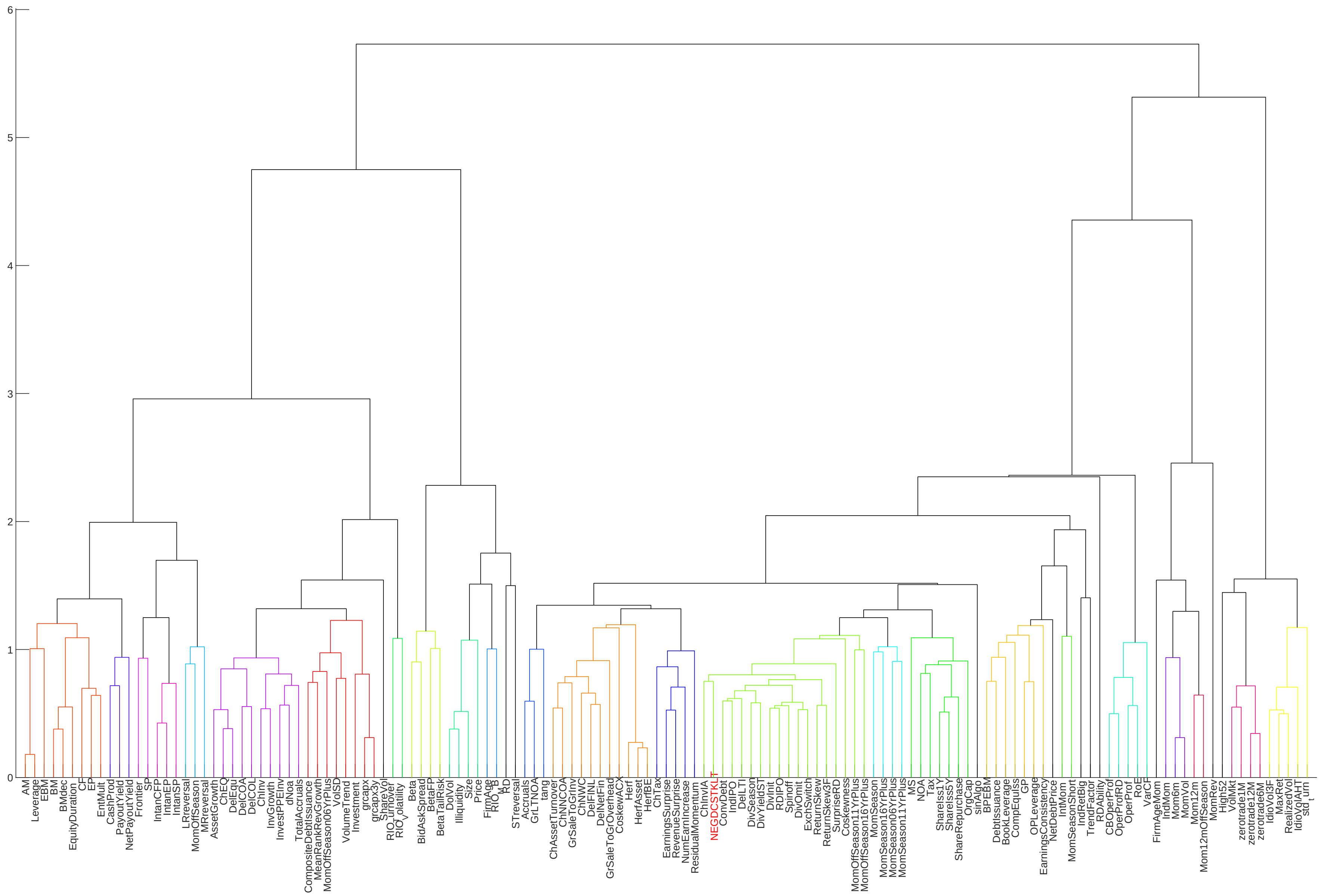


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

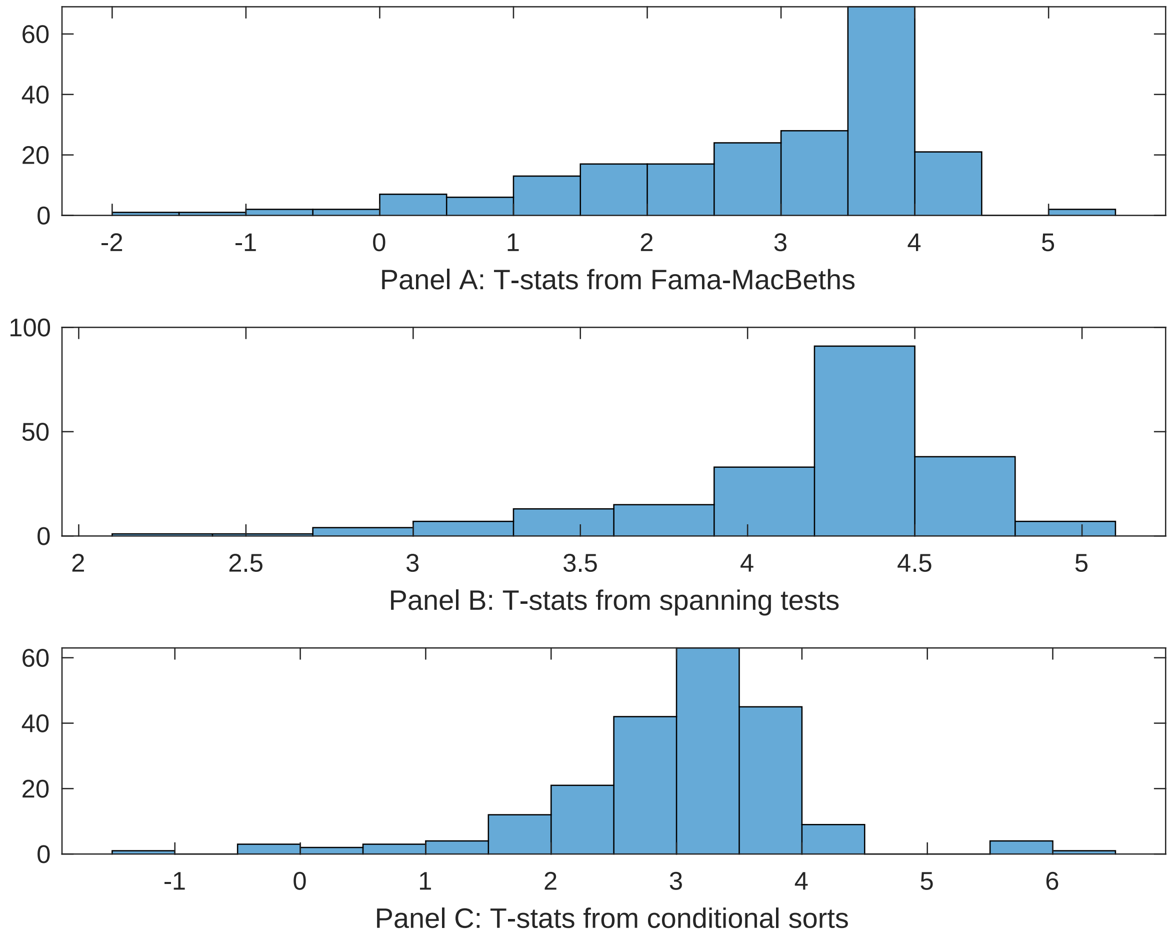


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of EDB conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDB} EDB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EDB. Stocks are finally grouped into five EDB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDB trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on EDB. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EDB} EDB_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13	0.12	0.17	0.13	0.12	0.13	0.12
	[6.06]	[5.42]	[6.92]	[6.45]	[5.70]	[6.16]	[3.95]
EDB	0.23	0.14	0.21	0.20	0.26	0.21	0.54
	[2.96]	[1.57]	[2.44]	[2.78]	[2.89]	[2.34]	[0.70]
Anomaly 1	0.14						0.69
	[6.13]						[2.50]
Anomaly 2		0.21					0.92
		[1.90]					[0.87]
Anomaly 3			0.40				-0.12
			[3.40]				[-0.53]
Anomaly 4				0.27			0.24
				[4.88]			[4.23]
Anomaly 5					0.11		-0.79
					[3.19]		[-0.13]
Anomaly 6						0.94	0.64
						[8.02]	[5.40]
# months	715	715	720	715	720	720	715
$\bar{R}^2(\%)$	0	1	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the EDB trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EDB} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.18	0.23	0.18	0.14	0.20	0.20	0.15
	[2.48]	[3.08]	[2.52]	[1.81]	[2.67]	[2.57]	[2.10]
Anomaly 1	27.79						12.84
	[7.38]						[2.85]
Anomaly 2		18.26					6.19
		[6.28]					[1.97]
Anomaly 3			34.18				36.74
			[8.60]				[6.40]
Anomaly 4				25.87			11.04
				[6.45]			[2.47]
Anomaly 5					17.31		-5.09
					[4.60]		[-0.96]
Anomaly 6						-1.57	-20.58
						[-0.34]	[-4.17]
mkt	1.63	3.77	2.06	2.12	0.63	0.95	3.97
	[0.92]	[2.05]	[1.18]	[1.18]	[0.35]	[0.52]	[2.25]
smb	1.29	3.18	-0.93	1.91	-0.01	0.34	3.03
	[0.50]	[1.20]	[-0.37]	[0.73]	[-0.00]	[0.13]	[1.20]
hml	4.99	-1.41	1.53	7.83	3.45	5.85	0.56
	[1.46]	[-0.38]	[0.45]	[2.26]	[0.98]	[1.65]	[0.16]
rmw	-9.57	-5.17	7.10	-3.79	7.06	5.50	-7.53
	[-2.38]	[-1.33]	[2.08]	[-1.00]	[1.99]	[1.54]	[-1.80]
cma	10.73	12.63	-5.14	15.41	10.54	29.88	3.96
	[1.94]	[2.25]	[-0.82]	[2.85]	[1.65]	[3.93]	[0.53]
umd	3.04	5.76	3.70	2.62	4.73	4.30	2.07
	[1.72]	[3.22]	[2.13]	[1.46]	[2.63]	[2.35]	[1.20]
# months	728	728	732	728	732	732	728
$\bar{R}^2$(%)	17	16	19	16	14	11	25

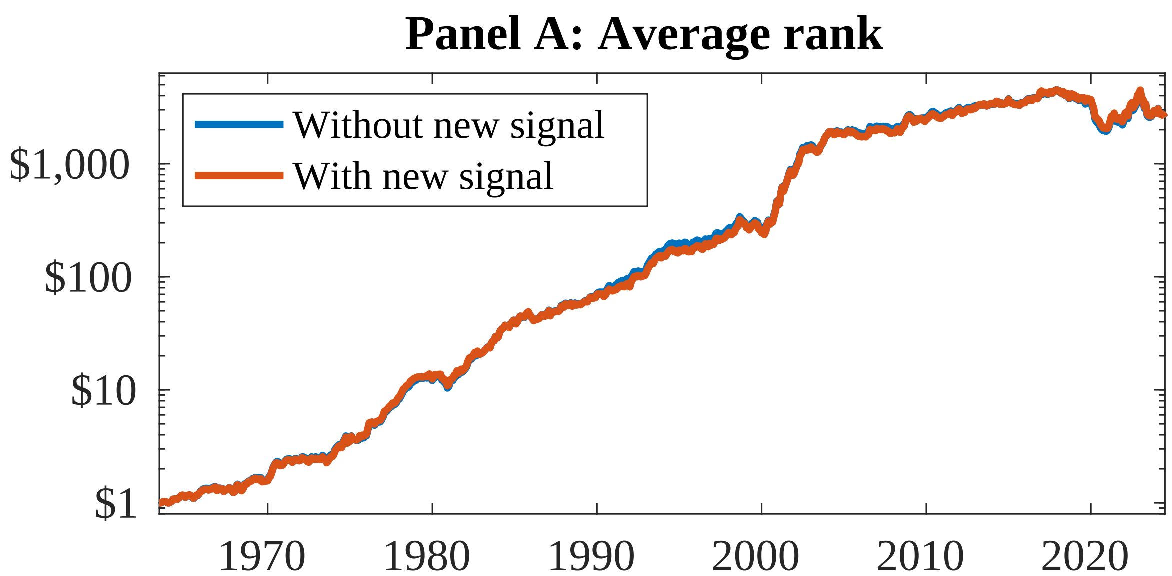


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EDB. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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